About this document

This document gives an overview of the topics you need to understand for the theory exam. There is no point in memorizing this document. This document merely lists all the topics, following the presentations, you need to understand and need to be able to explain and to answer insight questions about.

In the exam, a part of the questions are multiple choice (no guess correction) and a part of the questions are open questions.

The questions on the exam test if you understand the processes and protocols, they are not about memorizing specific details.

The following kind of questions will **not** be on the exam:

- Give all the fields of the IP header
- Give the port numbers of the DNS protocol
- Give the colors for an A-type ethernet connector
- Which port numbers in TCP and UDP are used as well-known ports
- Give 5 application layer protocols that use UDP

The following kinds of questions can be on the exam

- Which protocol defines the rate at which segments are send (TCP (=segments) using the Window field in the header)
- What is the ARP protocol used for? (Getting the MAC address of the device with a given IP address)
- Which protocol drops a packet if there is a bit error in the received data (Ethernet using the CRC tailer)
- Which of the following is a valid IPv6 address
    - 2001:CAFE:1:1::1 ok
    - FF80::1 ok
    - 2001:F524E:5557:2587:1547:AA44:52DD:EE44 ok
    - 2001:F52G:5557:2587:1547:AA44:52DD:EE44 G is not a hexadecimal number
    - 2001:F524:5557:2587:1547:AA44:52DD to few hexteds, only 7 given, 8 needed
- Which of the following host IP address are in the same network as 192.168.0.1/25
    - 192.168.0.0 not a host ip address
    - 192.168.0.10 ok
    - 192.168.0.200 not in the same network
- What is the destination MAC address of a packet on a specific location in a given network diagram
- Which packets are send and received by a client to receive an IPv4/IPv6 address (IPv4: DHCP discover, offer, request, ack; IPv6 Router Solicitation, Router Advertisement)
- Which transport layer protocol does DHCP use (UDP because TCP needs to setup a session)

01 – What is a network

Peer-to-peer network and Client-Server networks

Components in a network

- Devices
  - End devices
  - Intermediate devices
- Media
  - Copper
  - Fiber Optic
  - Wireless
- Services

You have to be able to read network diagrams

- Recognize Server, Computer, Router, Switch, Hub, Internet symbols

LAN and WAN networks

Circuit switched and Packet switched networks

Networks have to provide

- Fault tolerance
- Scalability

02 – Building a basic network

Recognize LAN (RJ45) port

Switch usage

IPv4 and IPv6 addresses (more in chapter 7, 8 and 9)

Binary to decimal and decimal to binary

Hexadecimal to decimal and decimal to hexadecimal

Ipconfig (windows) and ip a (Linux) command

Ping command

Loopback address

Firewall usage

Router usage

Subnet mask (more in chapter 7 and 8)

Default gateway

03 – Network protocols

What is a protocol?

Unicast – Multicast – Broadcast – Anycast

Network protocol scope

- Sender and Receiver (addressing)
- Encoding (file format)
- Encapsulation (packet format)
- Segmenting (packet size)
- Access method (prevent collisions)
- Flow control (congestion avoidance)
- Response timeout and retransmission (reliability vs speed)

Layers of the OSI model (in more detail in next chapters)

- Names
- Responsibilities
- Devices working in the layers
- Addresses used in the layers
- Protocols working in these layers (the ones that are described in later chapters)

DHCP (more in chapter 11)

DNS (more in chapter 11)

NAT (more in chapter 7)

Process of adding headers to the data by the OSI layers (more in next chapters)

Communication between devices on the same network/on a remote network (more in next chapters)

04 – Introduction to Cisco IOS

Don't need to know this for the theory exam

05 – Network access and the physical layer (layer 1)

Copper – Fiber – Wireless

Physical layer responsibilities

- Transmit streams of bits
- Physical layer components

Encoding methods

- Manchester encoding
- Non-return to zero encoding

Modulation: What is it and why is it used

- Asynchronous
- Synchronous

- Frequency Modulation
- Amplitude Modulation
- Phase Modulation

- Channels
- Bandwidth – Throughput – Goodput

Characteristics of copper cables

- EMI
- Crosstalk

- UTP – STP – Coax

- Reason for twists in a Twisted Pair cable

- Difference between CAT3, CAT5, CAT5e, CAT6

- Straight – Crossover Cable usage (not the color codes)

Characteristics of fiber cables

- Single mode fiber and Multimode fiber

Fiber versus Copper cables

06 – Datalink layer and Ethernet (Layer 2)

Responsibilities

- Recognize start of stream (frame) and end of stream (frame)
- Bit error detection (CRC tailer)
- Media access control
- Data encapsulation (Header and Tailer)
- Physical addressing (MAC addresses)

LLC and MAC sublayer

Ethernet header and tailer

- Start of frame
- Addressing
- Type
- Error Detection (CRC tailer)
- Frame stop

MAC Addresses (Physical addresses)

- Frame processing
- 48 bit addresses (OUI = first 24 bit – Vendor assigned = last 24 bit)
- Unicast MAC and Broadcast MAC

MAC versus IP

Address Resolution Protocol (ARP)

- ARP operation
- ARP table
- Arp -a and arp -d commands
- ARP problems
- ARP spoofing

Switchport operation

- Learning – Aging – Fooding – Selective Forwarding
- MAC table

Broadcast domain

07 – Network Layer

Network layer responsibilities

- Addressing (IP)
- Encapsulation (Header)
- **Routing**
- De-Encapsulation

Switch = Layer2 device, works with MAC addresses (MAC table)

Router = Layer3 device, works with IP addresses (Routing table)

IPv4 and IPv6 (More in chapter 8 and 9)

Characteristics

- Connectionless
- Best Effort
- Media Independent
- MTU

IPv4 Header

- Version, Differentiated Services (QoS), Time to Live, Protocol, Source and Destination IP

IPv4 address (More in chapter 8)

IPv4 address configuration

- Static
- DHCP

IPv4 Limitation

- IPv4 address depletion
- Internet routing table expansion
- No end to end connectivity (NAT)

Network Address Translation (NAT)

IPv6 (More in chapter 9)

IPv6 Header

- Version, Traffic Class, Next Header, Hop Limit, Source and Destination address

Host routing table

- Itself (127.0.0.1 or Localhost)
- Host on local network
- Host on remote network

Netstat -r and route print commands

Router routing table

- Directly connected networks, Static route, Dynamic route, Default route
- Administrative distance
- Next Hop

Reading routing table entry

Internet hierarchy

- Tier1, Tier2 and Tier3 networks
- Transit and peering

Router specifics + router configuration (starting from 71 to 87) not to know for the theory exam

08 – IPv4

IP address structure

Binary to decimal and decimal to binary

Subnet mask and prefix length (CIDR notation)

- Network portion – Host portion

Network address, host address, broadcast address

Private address blocks

- Usage
- 192.168.0.0/16, 172.16.0.0/12 and 10.0.0.0/8

Subnetting

- Subnet network (Based on required hosts or networks)
- Get Network and broadcast from given IP address
- Get number of hosts on a network
- VLSM principle (No exercises)

09 – Ipv6

Need for IPv6

Transition

IPv6 addresses

Rules to shorten IPv6 addresses

- Omit leading 0
- Omit all 0 segments

Prefix length

IPv6 address types

- Unicast
  - Link local
  - Global unicast
- Multicast
- Anycast

IPv6 address distribution principle (no prefix length numbers)

IPv6 address configuration

- Static
- DHCP
- SLAAC

Stateless Address Autoconfiguration (SLAAC)

- Router solicitation – router advertisement
- /64 interface ID generation (MAC or random)

Neighbor solicitation and Neighbor advertisement

- IP to MAC address (similar to ARP in IPv4)
- Duplicate address detection

Multicast principle

- Devices subscribe to multicast
- FF02::1 all nodes on subnet versus broadcast in IPv4

IPv6 header

Tracert command

IPv4 and IPv6 coexistance

- Dual stack
- Tunneling
- Translation

10 – Transport layer

Responsibilities

- Tracking (TCP sessions)
- Segmenting data (TCP sequence number)
- Identify applications (port numbers)

TCP vs UDP

TPC

- Establish session
- Reliable delivery (ACK)
- Same order delivery (Sequence number)
- Flow Control (window and TCP slowstart)

TCP header

- Source and Destination Port, Sequence Number, Ack number, Window

UDP

- Connectionless (No session)
- Unreliable delivery
- No order reconstruction
- No flow control

UPD header

- Source and Destination port

Port number usage and mechanism

Socket

Types of port numbers

- Well-known ports
- Registered ports
- Dynamic ports

Netstat command

TCP segmentation

Setting up (Three way handshake) and tearing down TCP session

TCP ordered delivery (Sequence numbers)

TCP reliability (Ack numbers)

Window size

UDP and TCP use cases

11 – Application layer

DNS

DHCP

NAT

NTP

HTTP

HTTPS

Nslookup command